



Web Developer Onboarding

General

Hi, welcome to the **Data Driven Education - Discussion Forums** subteam! This is a document with info so that you can get familiar with our progress thus far and make it easier to transition into the group.

The web development team is one of two sub-subteams, with the other being data science. Historically, there's been quite a disconnect between the two groups but we have been putting more emphasis on communication between the groups. Try to keep that in mind throughout the semester because without the data, the web-app is useless!

Tools

React.js for frontend web-development

- We haven't done much on this front, but it seems to be the easiest framework to connect with Flask.

- Some tutorials:

- <https://www.youtube.com/watch?v=w7ejDZ8SWv8>
 - <https://www.youtube.com/watch?v=cb1vy1HpVwk>
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Flask for backend web-development

- We are primarily using Flask because of its many, many libraries that handle data manipulation, storing data, serving content to web-app, and much more. It plays very well with **SQLite3**, which currently stores our anonymized discussion forum data from CS 6601.

- There is a Flask app in the GitHub called **database-app** that uses a database (data.db) to serve graphs (generated by **matplotlib**) to a local web server.

- Some tutorials:

- https://www.youtube.com/watch?v=Z1RJmh_OqeA
 - <https://www.youtube.com/watch?v=Qr4QMBUPxWo>
 - <https://www.youtube.com/watch?v=byHcYRpMgl4> (SQLite3)
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Figma for developing hi-fidelity prototypes and user testing

- Extremely easy to work on overall UI/UX, test different information layouts, and experiment with different ideas before jumping into coding. The past few semesters, we have put less of an emphasis on design because we want everything to be functional first and foremost. However, don't ignore it completely!

- <https://www.figma.com/>

- There should be a Figma file with a couple iterations of a high-fidelity prototype. This is a good starting point for thinking about how the whole app should be presented, especially the dashboard.

Current Process

data.db file → **SQLite3** database → **Python** processing, graphs → **Flask** serves web-app

- data.db file is not being regenerated or updated yet
 - still need to generate graphs for sentiment, peak posting times (of day)
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Next Steps

There's a lot of work to go! Here's just some of the main priority features we need to implement:

- connecting Piazza or edX to our database
 - fetch real-time data from live courses
- connecting Flask to React
 - we are currently using Flask for both backend and frontend which is working right now, but it might be better to use React as the main generator for the frontend side of the web-application
 - rethink our development stack as a whole?
- figuring out how the data science team will use the real-time Piazza/edX data and perform transformations, analysis, etc. and then transport that data to our web-app